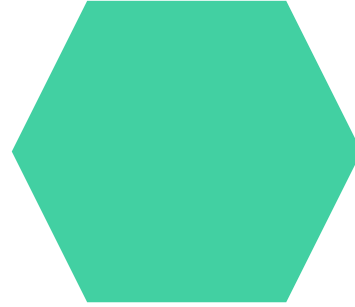
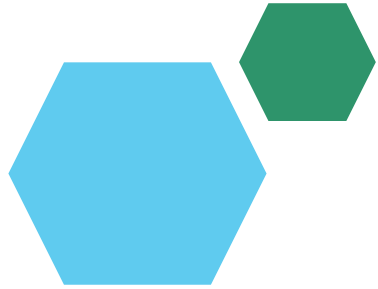




Project Title - Healthcare Analytics for Doctor Visits

Student Name: Charu yadav

College Name- Dr. Ram Manohar Lohiya Awadh University

AICTE STU ID : STU61b6174b3871e163932346



PROBLEM STATEMENT

- Healthcare providers and policymakers often assume that expanding insurance coverage automatically increases access to care. But it's unclear which factors actually drive how often people visit the doctor — is it insurance type, income, chronic health conditions, or demographics?
- This project analyses real health survey data to uncover what truly predicts doctor visits, and tests common assumptions about insurance and access against the evidence.



Project Description



This project analyses 5,190 individuals from an Australian health survey, examining the number of doctor visits alongside demographics, income, health status, insurance type, and chronic condition indicators. Using Python (Pandas, Seaborn, SciPy) in Jupyter Notebook, the data was cleaned, rescaled for interpretability, and explored through univariate, bivariate, and multivariate analysis.

Statistical tests (t-tests, chi-square) were used to confirm which patterns are real versus due to chance, followed by four student-designed deep-dive analyses into insurance effects and visit behaviour.



WHO ARE THE END USERS?



Healthcare policymakers designing insurance and access programs

Insurance providers evaluating who actually uses healthcare services

Public health researchers studying access and utilization patterns

Data analytics students studying real-world healthcare data



Technology Used

Python 3

Pandas & NumPy - data cleaning and analysis

Matplotlib & Seaborn - visualizations

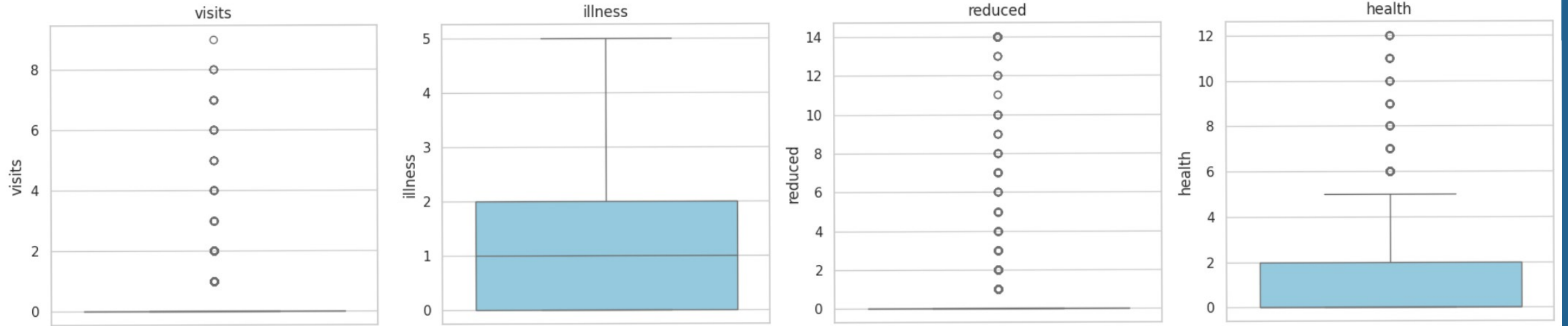
SciPy - statistical testing (t-tests, chi-square)

Jupyter Notebook

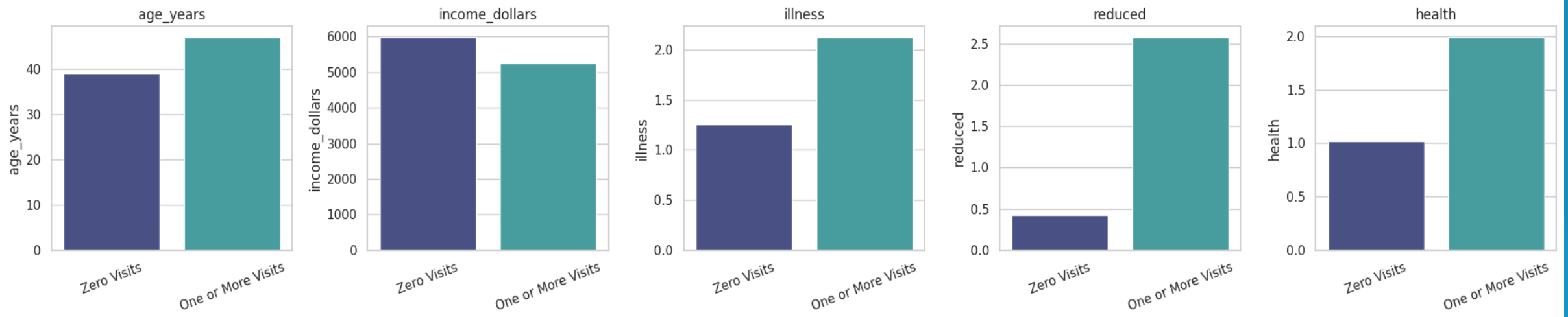


RESULTS

Outlier Investigation

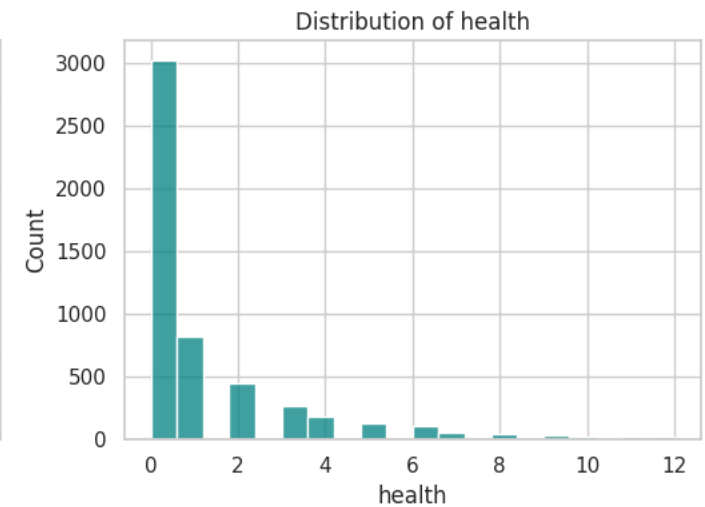
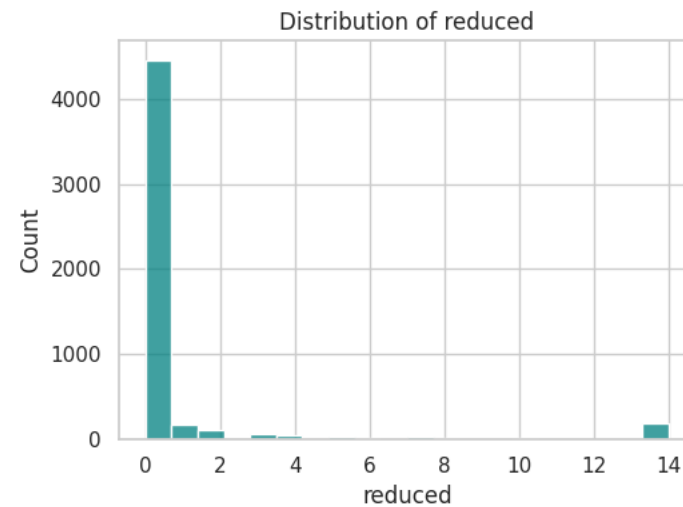
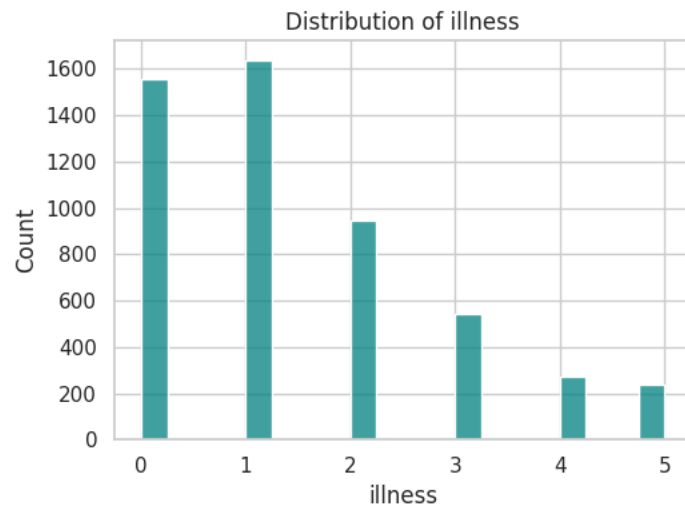
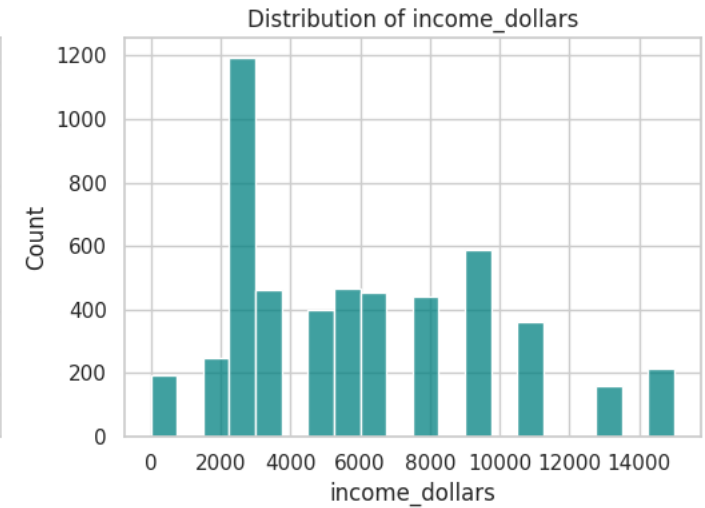
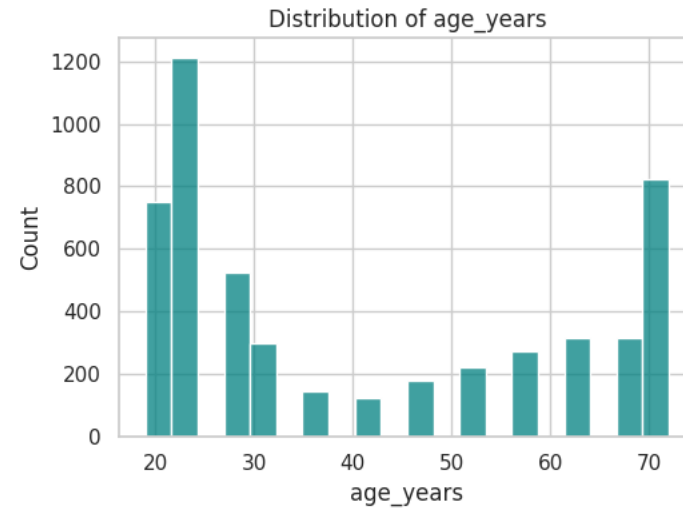
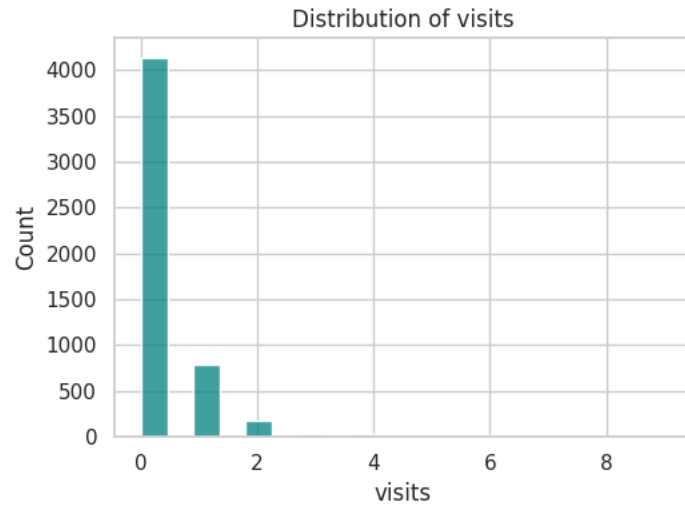


Ichronic vs any_visit



RESULTS

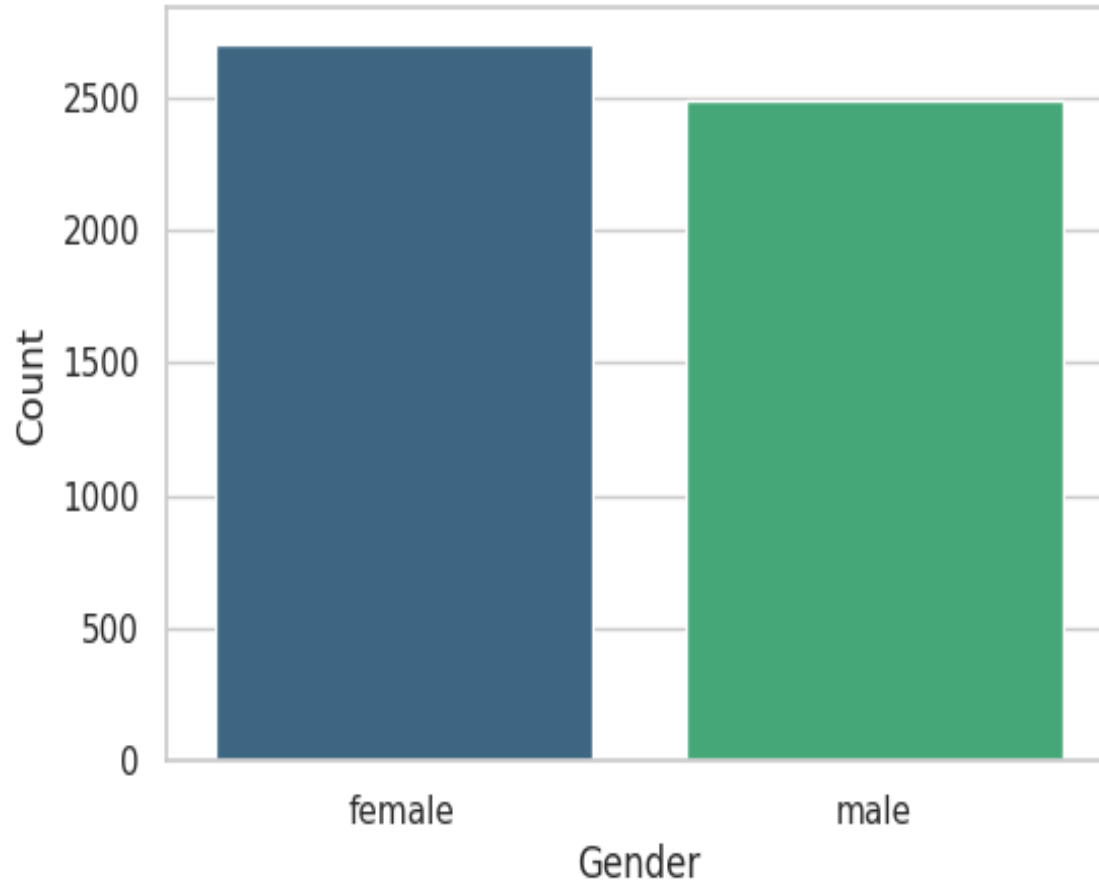
Univariate Analysis



RESULTS

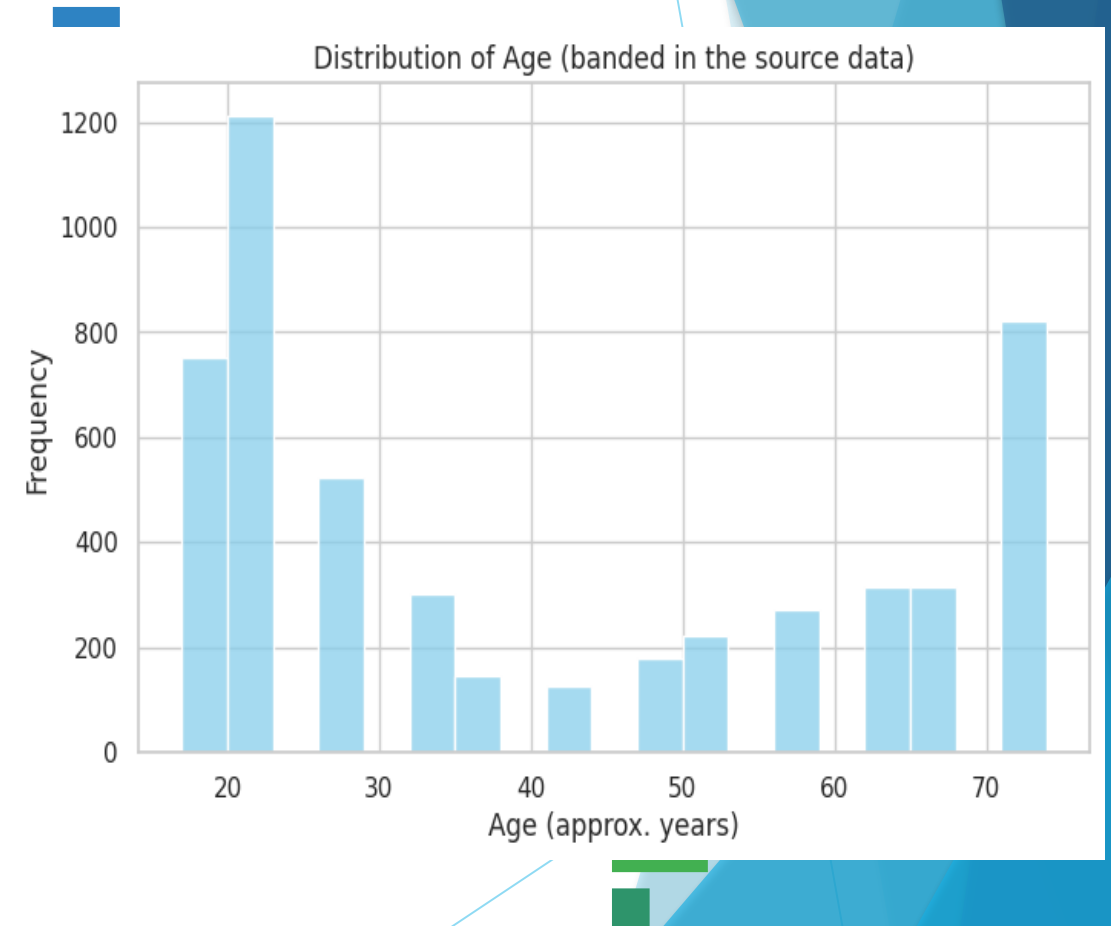
Distribution of Gender

Distribution of Gender



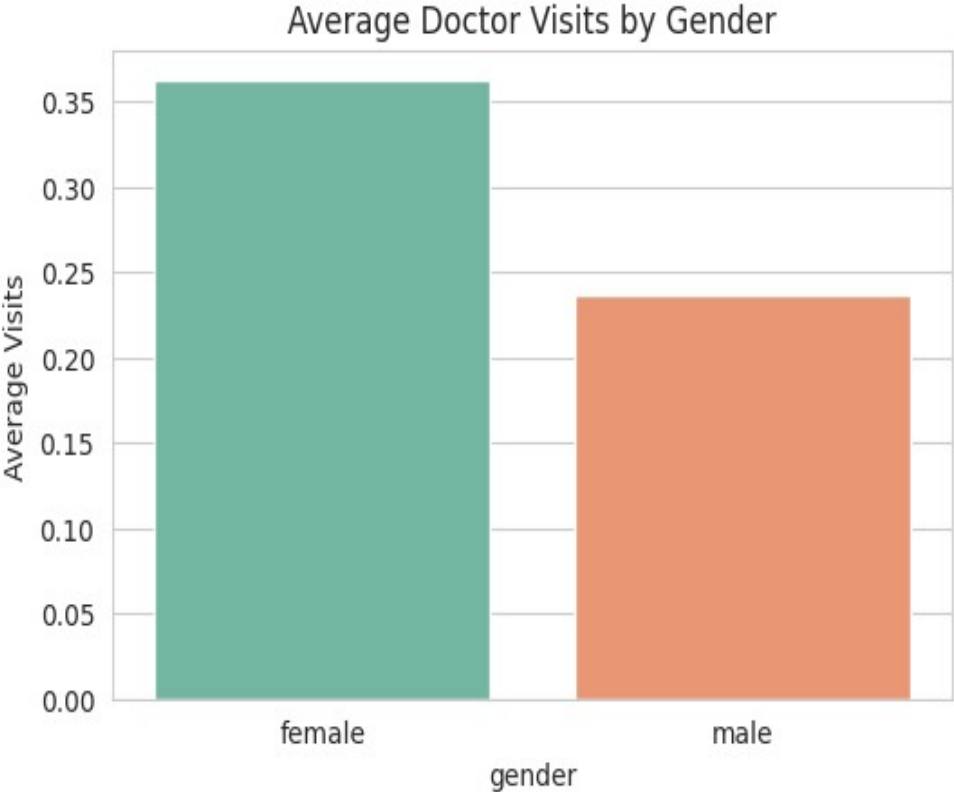
Distribution of Age

Distribution of Age (banded in the source data)

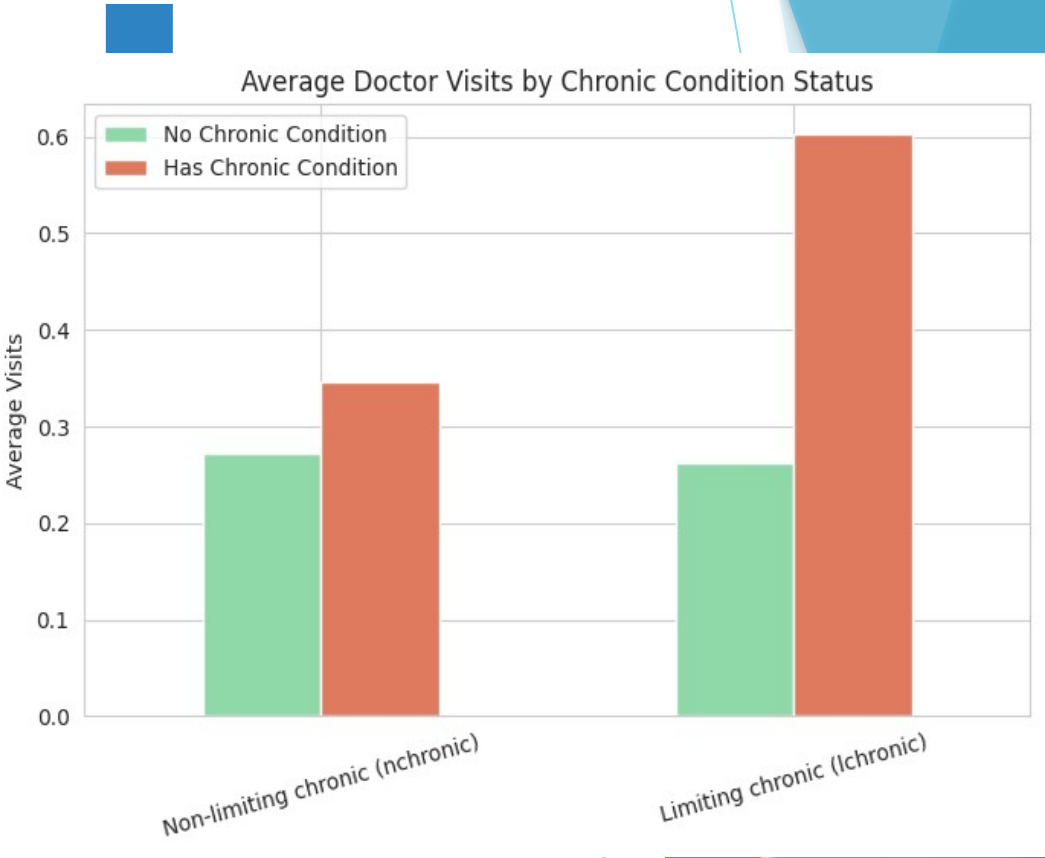


RESULTS

Visits by Gender



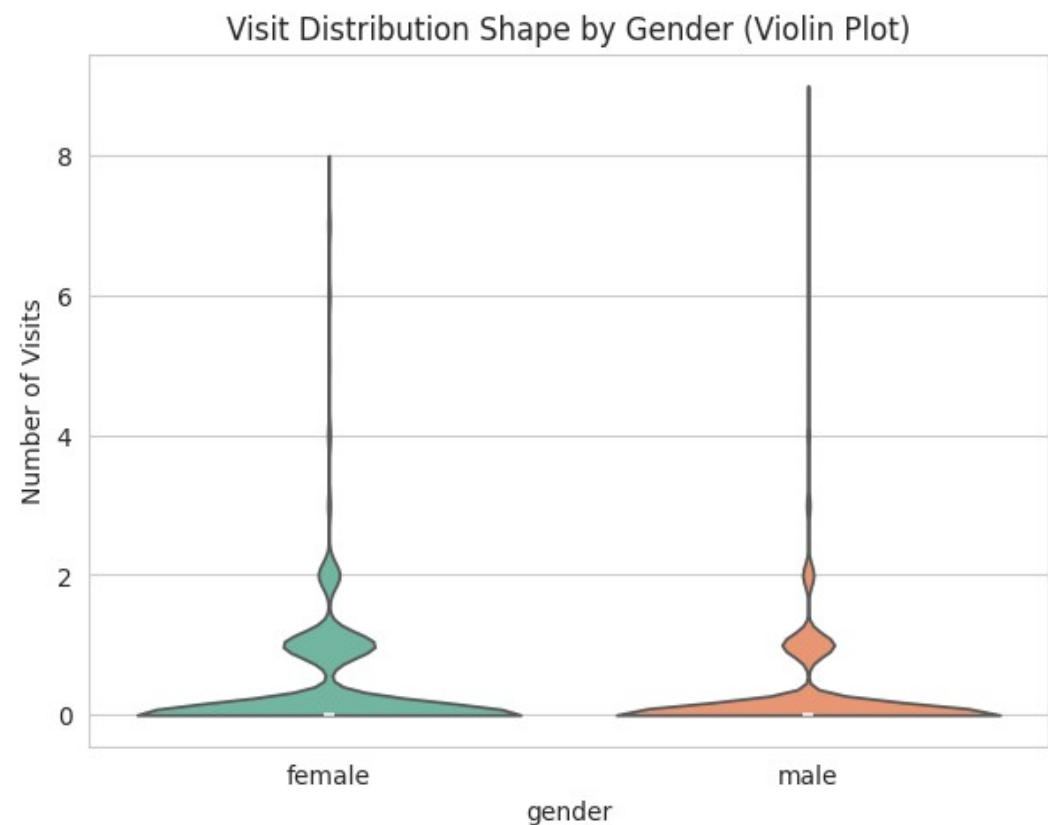
Chronic Condition



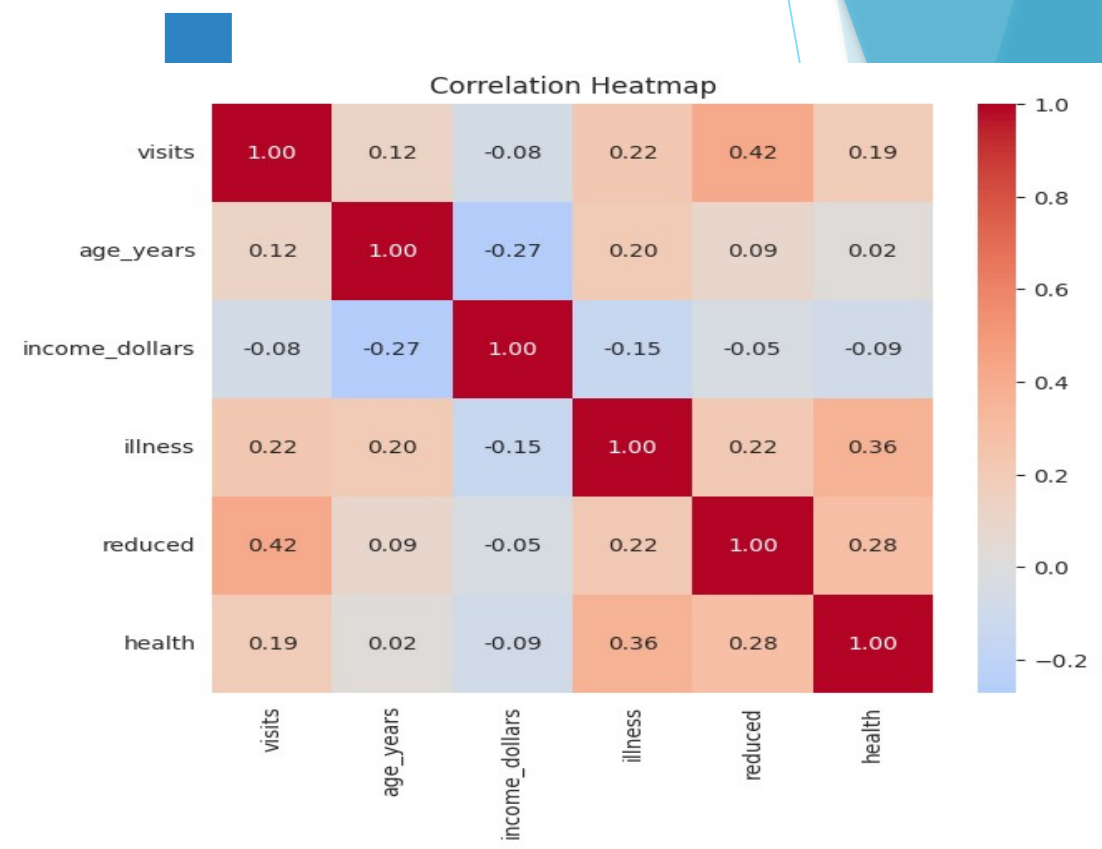
[Demo Link](#)

RESULTS

Visits by Gender (Violin)



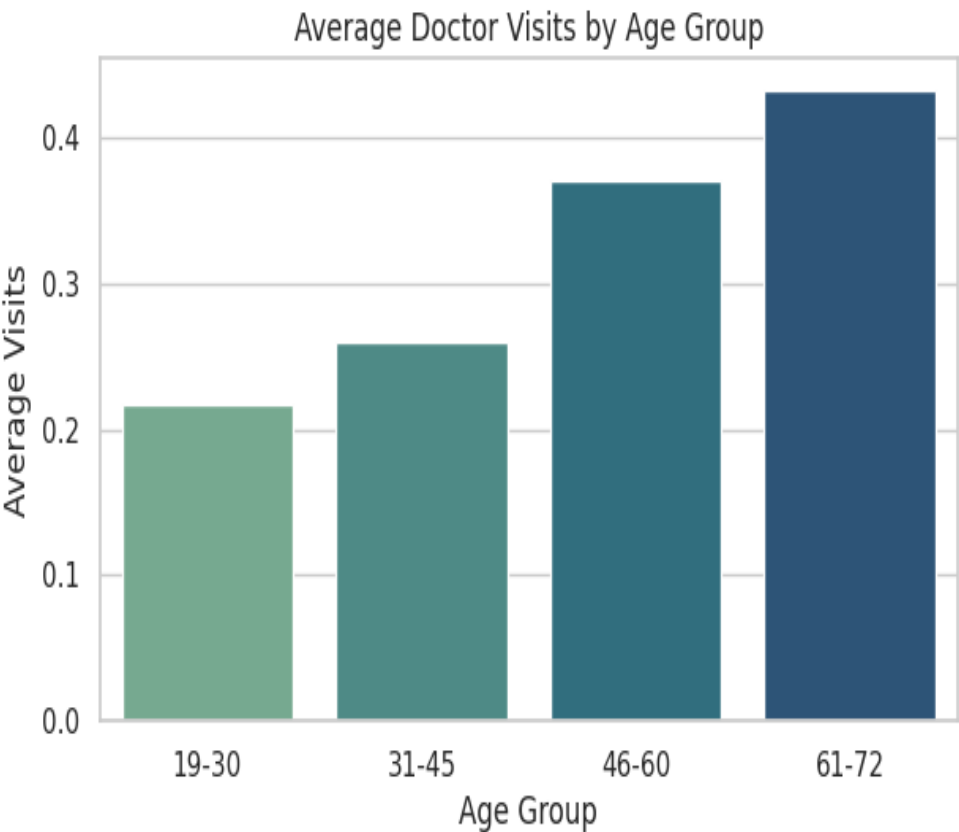
Correlation Heatmap



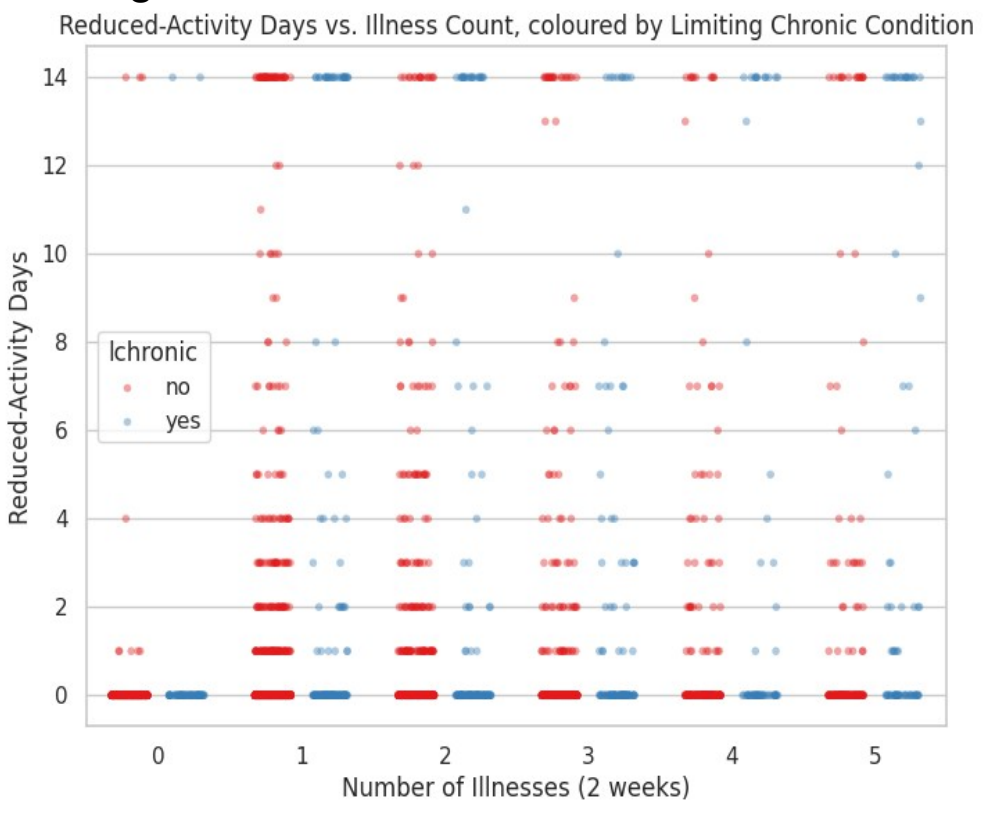
[Demo Link](#)

RESULTS

Doctor visits by age group



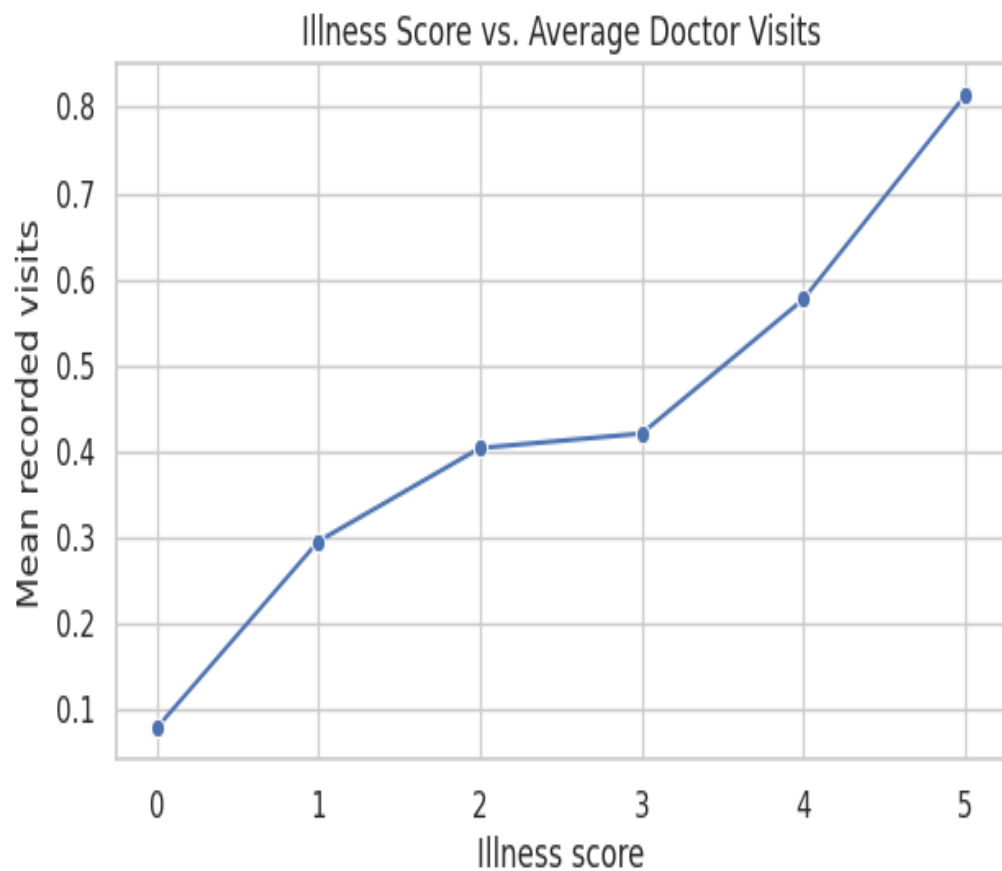
Chronic condition Age, visits and gender



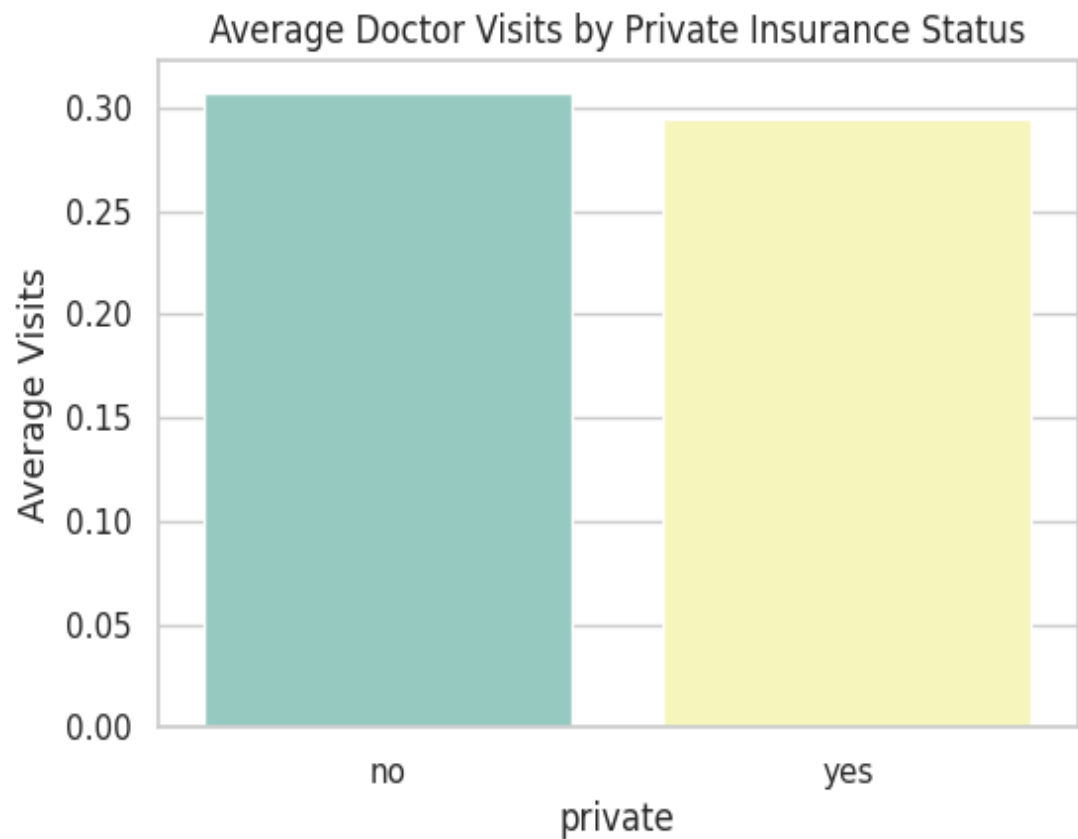
[Demo Link](#)

RESULTS

Illness score vs. average visits



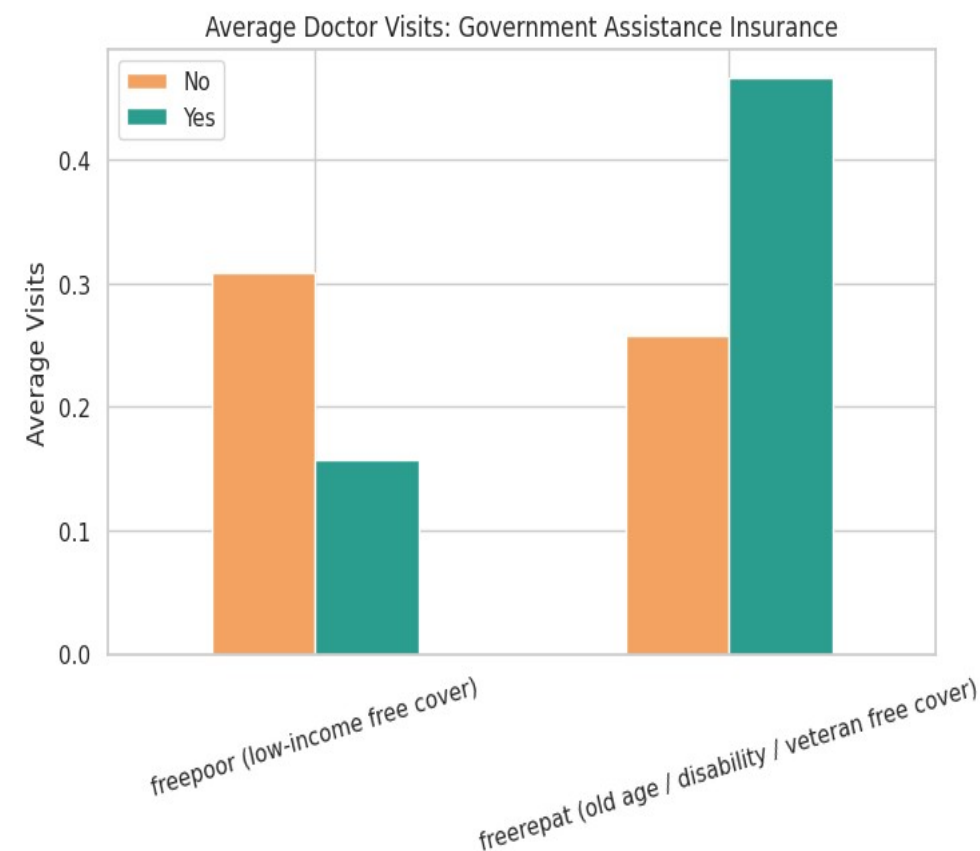
Private Insurance Increase Doctor Visits



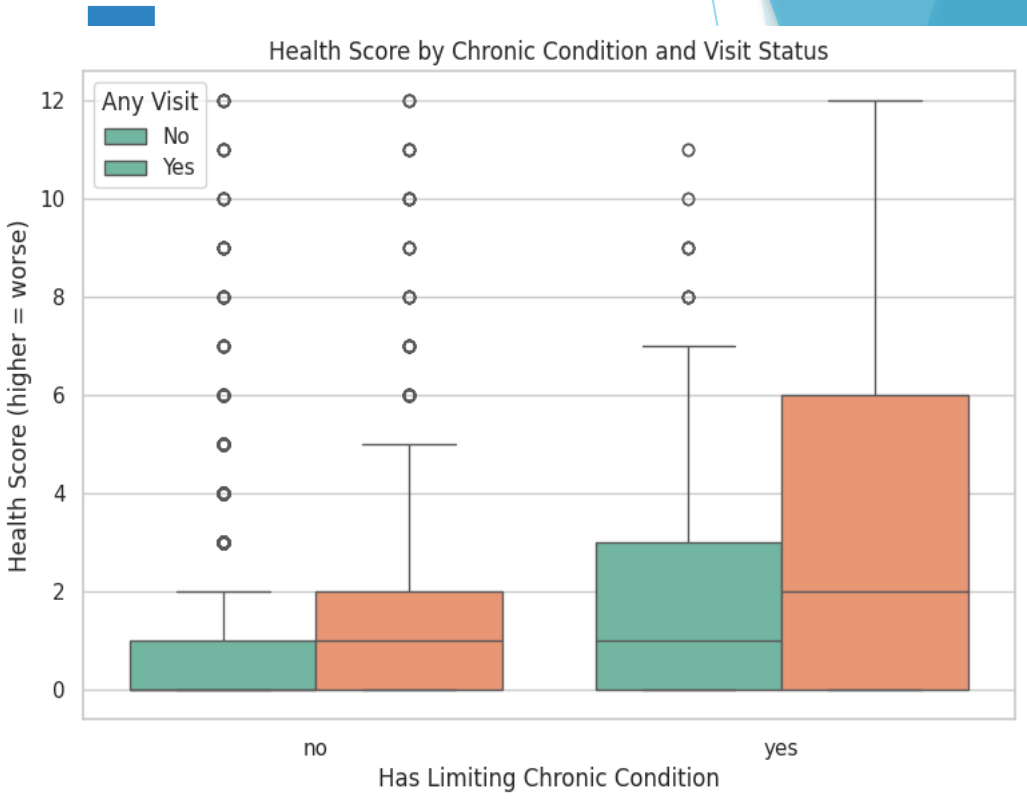
[Demo Link](#)

RESULTS

Government Assistance Insurance



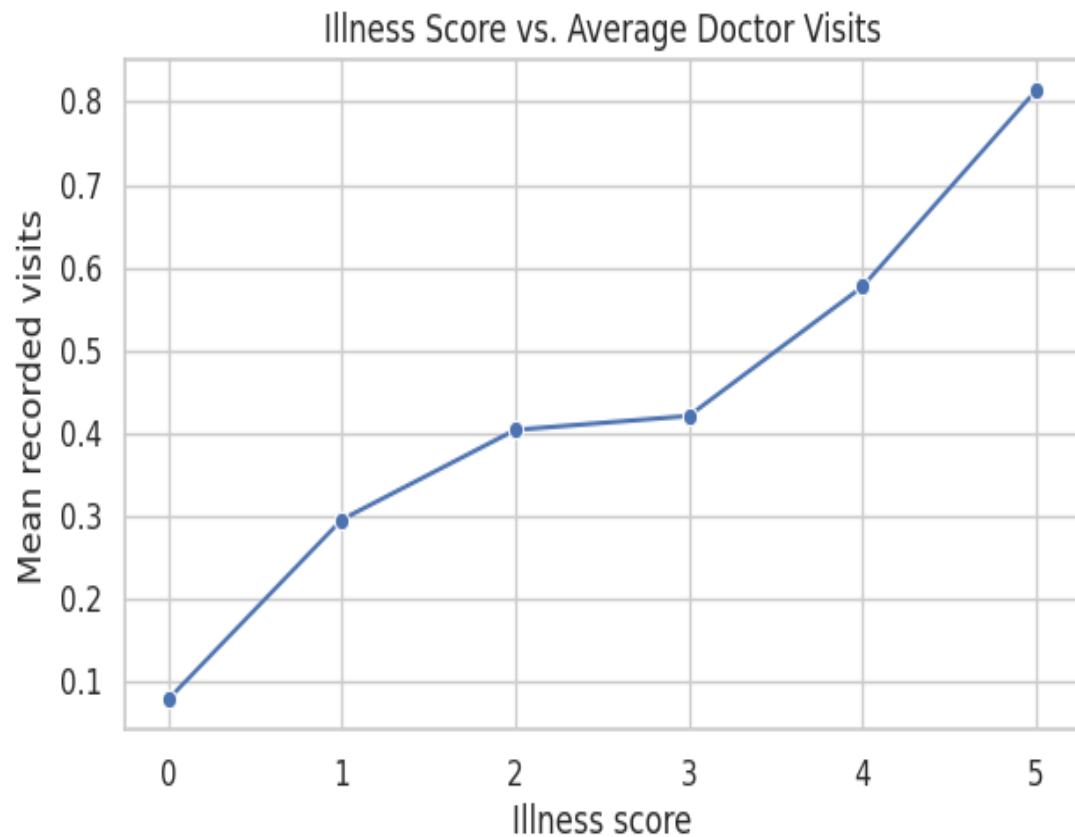
Health score by chronic condition and visit status



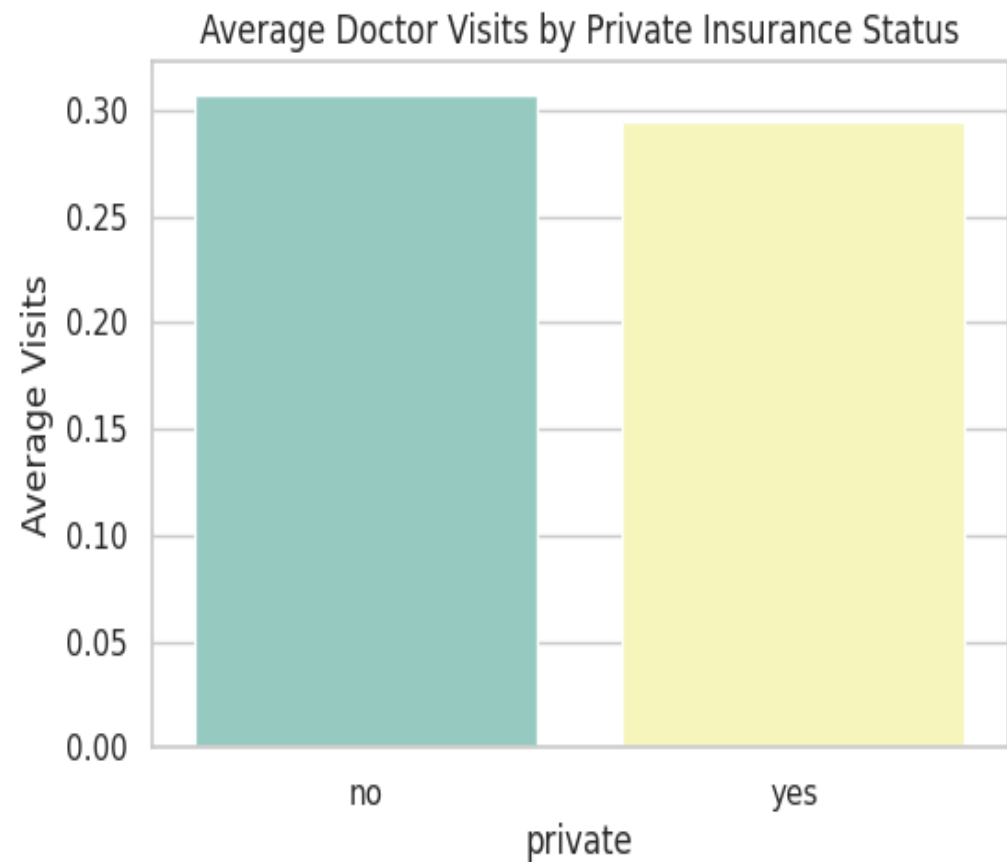
[Demo Link](#)

RESULTS

Illness score vs. average visits



Increase Doctor Visits



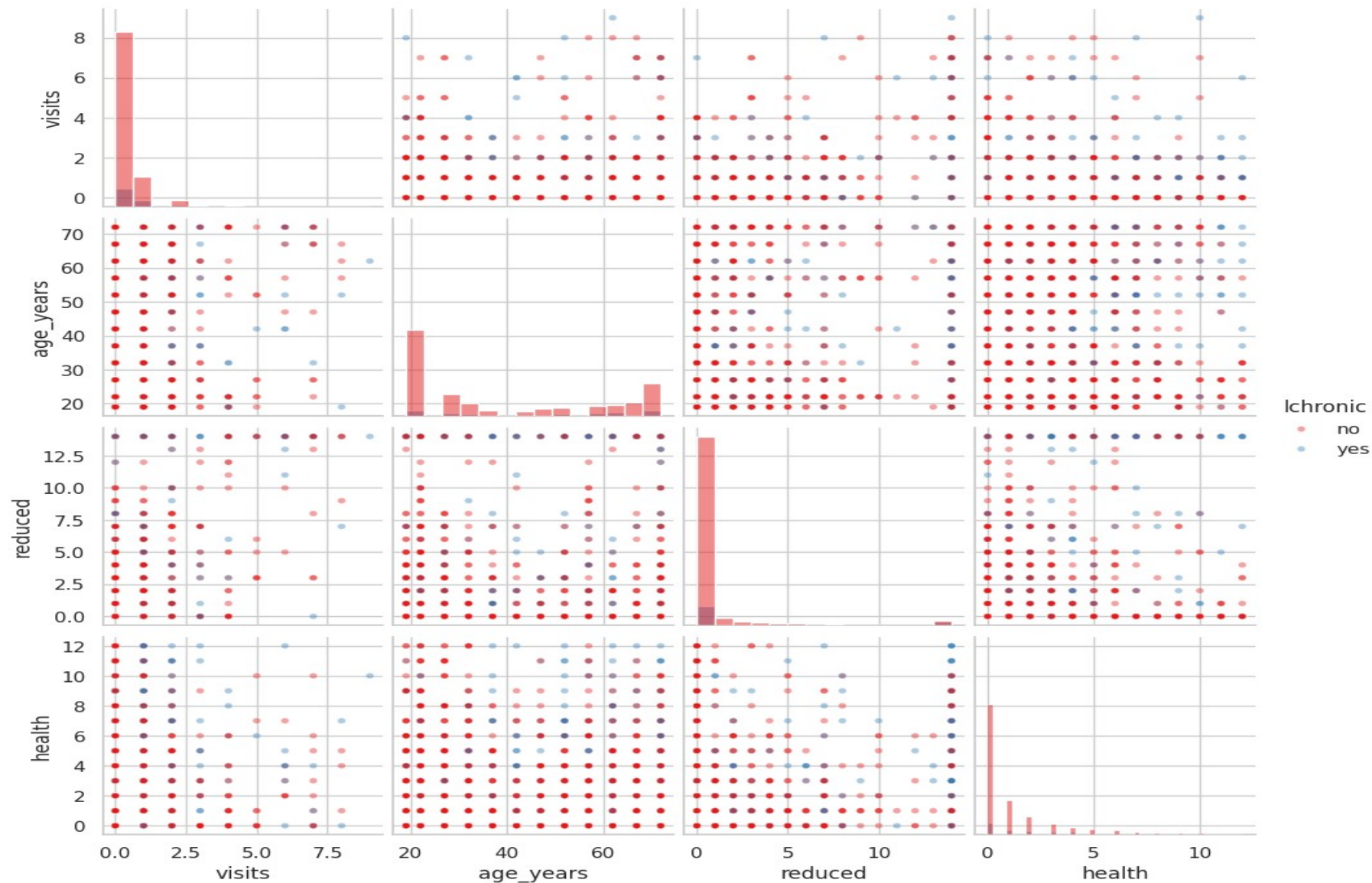
[Demo Link](#)

RESULTS

Multivariate Analysis:

Pair plot: visits, age, reduced-activity days, health, by chronic status

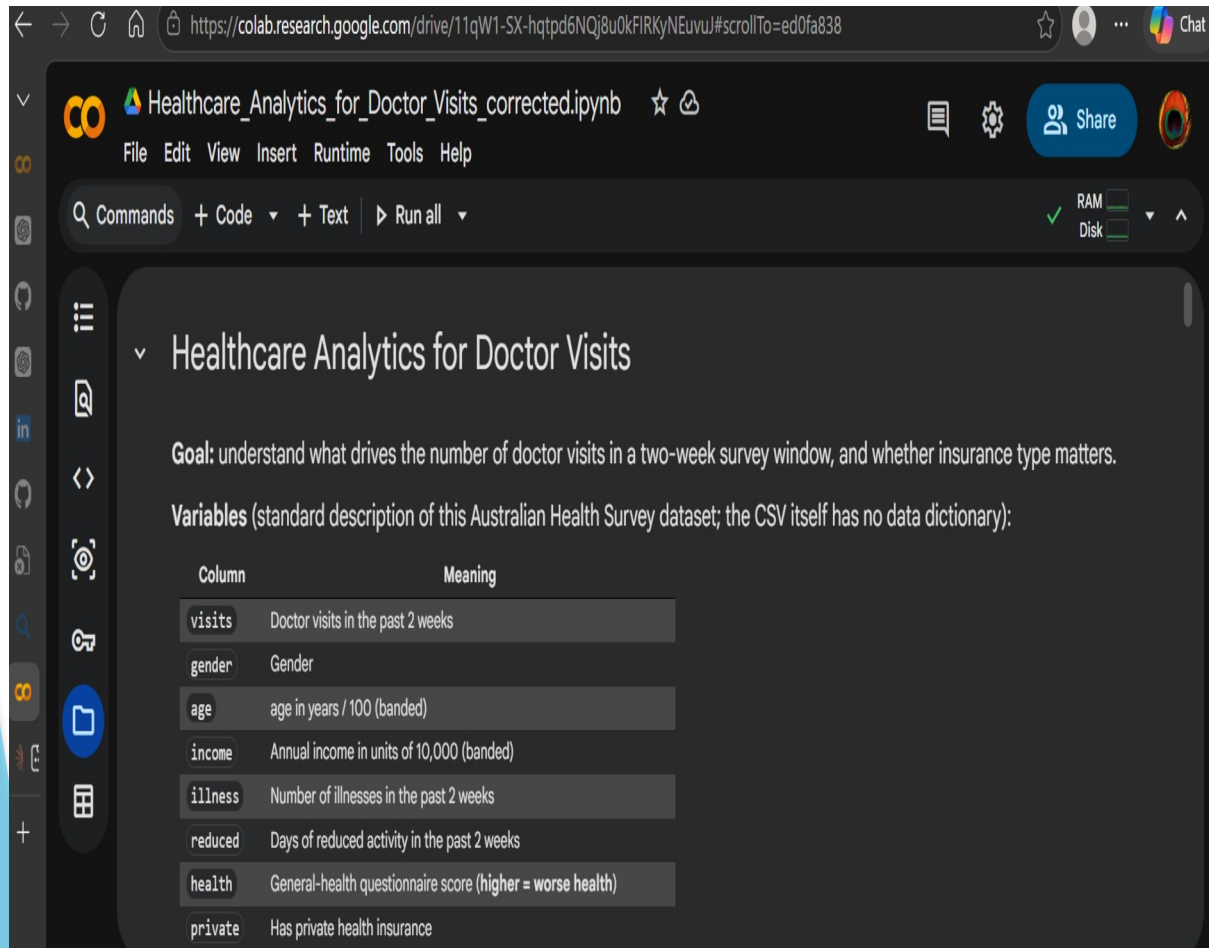
Multivariate Relationships: Visits, Age, Reduced-Activity Days & Health



[Demo Link](#)

RESULTS

Google colab and github screenshot



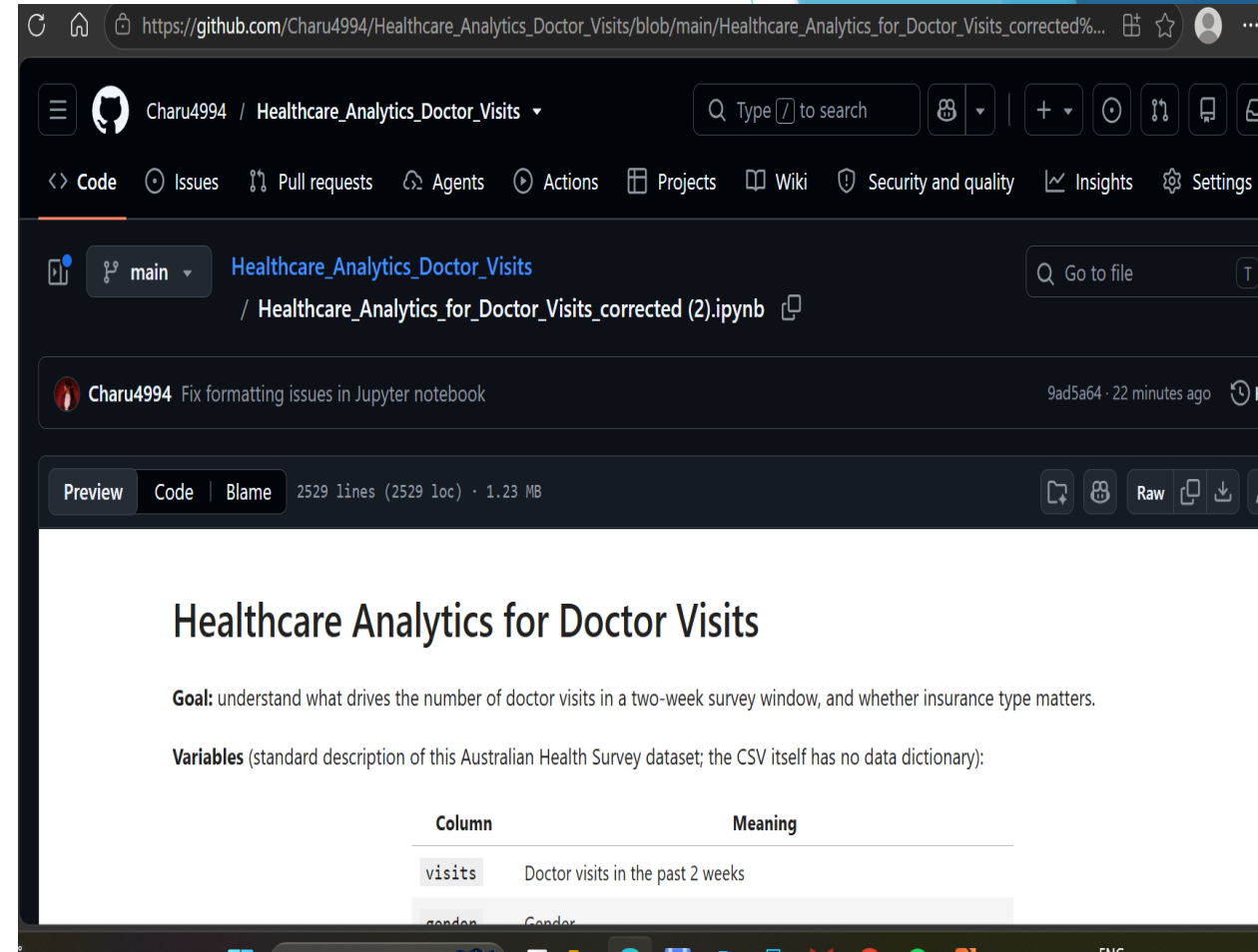
The screenshot shows a Google Colab notebook interface. The title bar indicates the file is 'Healthcare_Analytics_for_Doctor_Visits_corrected.ipynb'. The notebook content includes a title 'Healthcare Analytics for Doctor Visits', a goal statement, and a table of variables.

Healthcare Analytics for Doctor Visits

Goal: understand what drives the number of doctor visits in a two-week survey window, and whether insurance type matters.

Variables (standard description of this Australian Health Survey dataset; the CSV itself has no data dictionary):

Column	Meaning
visits	Doctor visits in the past 2 weeks
gender	Gender
age	age in years / 100 (banded)
income	Annual income in units of 10,000 (banded)
illness	Number of illnesses in the past 2 weeks
reduced	Days of reduced activity in the past 2 weeks
health	General-health questionnaire score (higher = worse health)
private	Has private health insurance



The screenshot shows a GitHub repository page for 'Charu4994 / Healthcare_Analytics_Doctor_Visits'. The repository contains a file named 'Healthcare_Analytics_for_Doctor_Visits_corrected (2).ipynb'. The file is shown in the 'Preview' view, displaying the same content as the Colab notebook screenshot.

Healthcare Analytics for Doctor Visits

Goal: understand what drives the number of doctor visits in a two-week survey window, and whether insurance type matters.

Variables (standard description of this Australian Health Survey dataset; the CSV itself has no data dictionary):

Column	Meaning
visits	Doctor visits in the past 2 weeks
gender	Gender

[Demo Link](https://github.com/Charu4994/Healthcare_Analytics_Doctor_Visits)

[:https://github.com/Charu4994/Healthcare_Analytics_Doctor_Visits](https://github.com/Charu4994/Healthcare_Analytics_Doctor_Visits)

Thank you

- Healthcare Analytics for Doctor Visits

- Demo Link: https://github.com/Charu4994/Healthcare_Analytics_Doctor_Visits

Charu Yadav